

# Quantum Fault Tolerance Meets Toom's Contours

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## 1 Shrinking

consider changing the name to something like hight lines / contour in order to hint on the similarity to topography map. In addition, the embedding inside  $\mathbb{R}^d$  should appears before that definition.

**Definition 1.1** (Potential Walls). We define the potential walls to be a functions collection from the vertices of  $\mathcal{G}_n^d$  to the reals as follow:

1. For any coordinate  $i \in [d]$ , we define  $L_i: \mathcal{G}_n^d \rightarrow \mathbb{R}$  to act as  $L_i(v) = v_i$ .
2. For any subset of coordinates  $I \subset [d]$  we define  $L_I: \mathcal{G}_n^d \rightarrow \mathbb{R}$  to act as  $L_I(v) = -\sum_{i \in I} v_i$ .

For a subset of vertices  $\Gamma \subset \mathcal{G}_n^d$ , a vertex  $v \in \Gamma$ , and a potential wall  $l: \mathcal{G}_n^d \rightarrow \mathbb{R}$ , we say that  $v$  is a *local maximum* of  $l$  in  $\Gamma$  if for any neighbor  $u$  of  $v$ , such that  $u \in \Gamma$ , we have  $l(v) \geq l(u)$ . We say that  $v$  is a strong local maximum if the inequality is strict, namely  $l(v) > l(u)$ . When the subset  $\Gamma$  is clear from the context, we might omit it and briefly say that  $v$  is a local maximum of  $l$ .

What we want is that if a bit contains both  $v, gv$  then there will be a check that contains both  $v, gv$ , that is true if  $d' > 1$ , then it means that we can erase from the bit a vertex which it is not  $gv$ .

**Claim 1.2.** Let  $z$  be an assignment, and let  $\partial z$  be the set of vertices adjacent to unsatisfied checks induced by  $z$ . Let  $v$  be a vertex in  $\partial z$ , and let  $g$  be a coordinate in  $S$  such that  $v$  is a local maximum of  $L_g$ . Then  $gv \notin \partial z$ .

*Proof.* Suppose  $gv \in \partial z$ , then

$$L_g(gv) = (gv)_g = 1 + v_g = 1 + L_g(v)$$

In contradiction to  $v$  being a local maximum of  $L_g$  in  $\partial z$ . □

**Claim 1.3.** Let  $z$  be an assignment, and let  $\partial z$  be the set of vertices adjacent to unsatisfied checks induced by  $z$ . Let  $v$  be a vertex in  $\partial z$ , and let  $g$  be a coordinate in  $S$  such that  $v$  is a local maximum of  $L_g$ . Then for any check associated with a  $d' - 1$  face rooted at  $v$  that contains  $gv$ , it is satisfied.

*Proof.* Assume, through contradiction, the existence of an unsatisfied check associated with a  $d' - 1$  positive face rooted at  $v$ , which contains  $gv$ . This implies  $gv \in \partial z$ , contradicting Claim 1.2.  $\square$

**Definition 1.4** (Small Code at Vertex  $v$ ). For a vertex  $v$ , we define the  $X$ -small code at vertex  $v$  to be the binary assignments over the bits ( $d'$  faces) rooted at  $v$  that satisfy  $\mathcal{C}_X$ 's restrictions rooted at  $v$  ( $d' - 1$  faces). Similarly, we define the  $Z$ -small code at vertex  $v$  to be the binary assignments over the bits ( $d'$  faces) rooted at  $v$  that satisfy  $\mathcal{C}_Z$ 's restrictions rooted at  $v$  ( $d' + 1$  faces). We denote by  $\mathcal{C}_v^-$  and  $\mathcal{C}_v^+$  the  $X$ -small code at  $v$  and  $Z$ -small code at  $v$ .

**Definition 1.5.** Should rewrite it better. We define  $\text{Stab}_v^-$  and  $\text{Stab}_v^+$  as the following linear codes:

$$\text{Stab}_v^- := \left\{ z : \partial z = 0 \text{ and } z \in \text{Span} \left( \{ \theta_{\mathbf{v}, \mathbf{S}'/\mathbf{g}}, \theta_{\mathbf{g}\mathbf{v}, \mathbf{S}'/\mathbf{g}} \text{ for } S' \subset S, |S'| = d' + 1, g \in S' \} \right) \right\}$$

$$\text{Stab}_v^+ := \left\{ z : \partial^* z = 0 \text{ and } z \in \text{Span} \left( \{ \theta_{\mathbf{v}, \mathbf{S}'/\mathbf{g}}, \theta_{\mathbf{g}\mathbf{v}, \mathbf{S}'/\mathbf{g}} \text{ for } S' \subset S, |S'| = d' + 1, g \in S' \} \right) \right\}$$

[COMMENT] Regarding the beneath, The number of restriction in  $\partial^+$  case was counted in negligently. In addition, we should show that for the  $\partial^-$  case, restrictions of the form  $\theta_{\mathbf{g}_2\mathbf{g}_1\mathbf{v}, \mathbf{S}}$  are degenerate. In the bottom line it's look like:

$$\partial^+ \theta_{\mathbf{g}_2\mathbf{g}_1\mathbf{v}, \mathbf{S}} = \partial^+ \theta_{\mathbf{g}_1\mathbf{v}, (\mathbf{S}/\mathbf{g}_3) \cup \mathbf{g}_2} + \partial^+ \theta_{\mathbf{g}_2\mathbf{v}, (\mathbf{S}/\mathbf{g}_3) \cup \mathbf{g}_1}$$

Lets develop it, consider a restriction rooted at  $g_2g_1v$ , and let's look on it's boundary:

$$\partial^+ \theta_{\mathbf{g}_2\mathbf{g}_1\mathbf{v}, \mathbf{S}'} = \sum_{g \in S/S'} \theta_{\mathbf{g}_2\mathbf{g}_1\mathbf{v}, \mathbf{S}' \cup \mathbf{g}} + \sum_{g \in S/S'} \theta_{\mathbf{g}^{-1}\mathbf{g}_2\mathbf{g}_1\mathbf{v}, \mathbf{S}' \cup \mathbf{g}}$$

Now, any term in the left closure is a bit beyond the scope of the bits in  $\text{Stab}_v^-$ , thus, we are zeroing them out. The bits, on the right closers are in the scope, only if  $g \in \{g_1, g_2\}$ . Thus, in total the check is equivalence to:

$$\begin{aligned} \partial^+ \theta_{\mathbf{g}_2\mathbf{g}_1\mathbf{v}, \mathbf{S}'} &= \theta_{\mathbf{g}_1\mathbf{v}, \mathbf{S}' \cup \mathbf{g}_2} \cdot \mathbf{1}_{g_2 \notin S'} \\ &\quad + \theta_{\mathbf{g}_2\mathbf{v}, \mathbf{S}' \cup \mathbf{g}_1} \cdot \mathbf{1}_{g_1 \notin S'} \\ \partial^+ \theta_{\mathbf{g}_1\mathbf{v}, (\mathbf{S}'/\mathbf{g}_3) \cup \mathbf{g}_2} &= \sum_{g \in S/S'} \theta_{\mathbf{g}_1\mathbf{v}, (\mathbf{S}'/\mathbf{g}_3) \cup \mathbf{g}_2 \cup \mathbf{g}} + \sum_{g \in S/S'} \theta_{\mathbf{g}^{-1}\mathbf{g}_1\mathbf{v}, (\mathbf{S}'/\mathbf{g}_3) \cup \mathbf{g}_2 \cup \mathbf{g}} \\ \partial^+ \theta_{\mathbf{g}_2\mathbf{v}, (\mathbf{S}/\mathbf{g}_3) \cup \mathbf{g}_1} &= \sum_{g \in S/S'} \theta_{\mathbf{g}_2\mathbf{v}, (\mathbf{S}'/\mathbf{g}_3) \cup \mathbf{g}_1 \cup \mathbf{g}} + \sum_{g \in S/S'} \theta_{\mathbf{g}^{-1}\mathbf{g}_2\mathbf{v}, (\mathbf{S}'/\mathbf{g}_3) \cup \mathbf{g}_1 \cup \mathbf{g}} \end{aligned}$$

$$\begin{aligned}\partial^+ \theta_{\mathbf{g}_1 \mathbf{v}, (\mathbf{S}'/\mathbf{g}_3) \cup \mathbf{g}_2} &= \left( \sum_{g \in S/S'} \theta_{\mathbf{g}_1 \mathbf{v}, (\mathbf{S}'/\mathbf{g}_3) \cup \mathbf{g}_2 \cup g} \right) + \theta_{\mathbf{v}, (\mathbf{S}'/\mathbf{g}_3) \cup \{\mathbf{g}_2, \mathbf{g}_1\}} \\ \partial^+ \theta_{\mathbf{g}_2 \mathbf{v}, (\mathbf{S}/\mathbf{g}_3) \cup \mathbf{g}_1} &= \left( \sum_{g \in S/S'} \theta_{\mathbf{g}_2 \mathbf{v}, (\mathbf{S}'/\mathbf{g}_3) \cup \mathbf{g}_1 \cup g} \right) + \theta_{\mathbf{v}, (\mathbf{S}'/\mathbf{g}_3) \cup \{\mathbf{g}_1, \mathbf{g}_2\}}\end{aligned}$$

Other way:

$$\begin{aligned}\partial^+ \sum_{g' \in S/S'} \theta_{\mathbf{g}' \mathbf{v}, \mathbf{S}'} &= \sum_{g' \neq g \in S/S'} \theta_{\mathbf{g}' \mathbf{v}, \mathbf{S}' \cup g} + \sum_{g \in S/S'} \theta_{\mathbf{v}, \mathbf{S}' \cup g} \\ &= \sum_{g' \neq g \in S/S'} \theta_{\mathbf{g}' \mathbf{v}, \mathbf{S} \cup g} + \partial^+ \theta_{\mathbf{v}, \mathbf{S}'}\end{aligned}$$

Where we use the fact that if  $g = g'$  then the bit is beyond the scope. In addition:

$$\begin{aligned}\partial^+ \sum_{g' < g \in S/S'} \theta_{\mathbf{g}' \mathbf{g} \mathbf{v}, \mathbf{S}'} &= \sum_{g' < g \in S/S'} \theta_{\mathbf{g}' \mathbf{v}, \mathbf{S}' \cup g} + \theta_{\mathbf{g} \mathbf{v}, \mathbf{S}' \cup g'} \\ &= \sum_{g' \neq g \in S/S'} \theta_{\mathbf{g}' \mathbf{v}, \mathbf{S}' \cup g} \\ &= \partial^+ \sum_{g' \in S/S'} \theta_{\mathbf{g}' \mathbf{v}, \mathbf{S}'} + \partial^+ \theta_{\mathbf{v}, \mathbf{S}'}\end{aligned}$$

Thus, we find more  $\binom{d}{d'-1}$  restriction over the checks. [We need to show that those restrictions are independent]. In addition:

$$\partial^+ \sum_{S'} \sum_{g' < g \in S/S'} \theta_{\mathbf{g}' \mathbf{g} \mathbf{v}, \mathbf{S}'} = 0$$

$$\partial^+ \theta_{\mathbf{g}_2 \mathbf{g}_1 \mathbf{v}, \mathbf{S}} = \partial^+ \theta_{\mathbf{g}_1 \mathbf{v}, (\mathbf{S}/\mathbf{g}_3) \cup \mathbf{g}_2} + \partial^+ \theta_{\mathbf{g}_2 \mathbf{v}, (\mathbf{S}/\mathbf{g}_3) \cup \mathbf{g}_1}$$

I don't succes to advance the idea more than that, So in the end I believe that I will do is to say that the diminsion of  $\text{Stab}_v^-$  is at least  $\binom{d}{d'+1}$ . And by fixing coordinates we add more  $\binom{d}{d'}$  restriction. So the dimension becomes:

$$\begin{aligned}&\binom{d}{d'+1} - \binom{d}{d'} \\ &\binom{d}{d-d'+1} - \binom{d}{d-d'}\end{aligned}$$

Note that since we talk on mapping checks  $\rightarrow$  bits, we use the  $\partial^+$ . Eventually, we need to use the poincare duality  $H^{d'} \rightarrow H_{d-d'}$ , the duality also should be used to define the SWEEP rule to  $C_Z$ . I didn't successes to do it other way.

**Claim 1.6.** Let  $v$  be a vertex. For any  $z \in \mathcal{C}_v^\pm$ , there is a stabilizer  $w \in \text{Stab}_v^\pm$  such that  $z = w|_v$ ; namely,  $z$  and  $w$  agree on the faces rooted at  $v$ .

*Proof.* We will show that codes obtained from  $\text{Stab}_v^\pm$  by forcing the faces rooted at  $v$  to  $z|_v$  have non-zero dimensions. Then it follows that any code word of  $\mathcal{C}_v^\pm$  could be extend to codewords in  $\text{Stab}_v^\pm$ . The number of bits in both  $\text{Stab}_v^\pm$  is:

$$\binom{d}{d'} + d \binom{d-1}{d'}$$

The numbers of checks for  $\text{Stab}_\pm$  are:

$$\binom{d}{d' \pm 1} + d \binom{d-1}{d' \pm 1}$$

The restriction to  $z|_v$  sets the bits rooted at  $v$  and satisfies the checks that are rooted at  $v$ , so the checks of the form  $\theta_{\mathbf{v}, S'}$  for  $|S'| = d' \pm 1$  are degenerate, and the dimension that remains is:

$$\dim \geq d \binom{d-1}{d'} - d \binom{d-1}{d' \pm 1}$$

In the case that  $d' = (d-1)/2$ , then for any  $x \in [d-1]$  we have  $\binom{d-1}{d'} > \binom{d-1}{x}$  therefore:

$$\begin{aligned} \dim &\geq d \binom{d-1}{d/2} - d \binom{d-1}{d/2 \pm 1} \\ &> d \end{aligned}$$

So there is at least  $2^d$  stabilizers  $w \in \text{Stab}_v$  that agree with  $z$ . □

**Claim 1.7.** Let  $z$  be a finite assignment. Suppose that

$$\max \{L_{d+1}(u) : u \in z\} > \max \{L_{d+1}(u) : u \in \partial^\pm z\}$$

Then there is  $\tilde{z} \sim_{\mathcal{C}_X} z$  such that:

$$\max \{L_{d+1}(u) : u \in \tilde{z}\} \leq \max \{L_{d+1}(u) : u \in z\} - 1$$

*Proof.* Let  $v_1, v_2, \dots, v_k$  be the vertices in  $z$  that achieve the maximum for  $L_{d+1}$ , namely  $v_1, v_2, \dots, v_k = \arg \max_{v \in z} L_{d+1}(v)$ , and assume that  $v_i \notin \partial^\pm z$  for any  $i \in [k]$ .

Consider the  $i$ th vertex  $v_i$ . By definition, since  $v_i \notin \partial^\pm z$ , we have  $\partial^\pm z|_{v_i} = 0$ , and since  $v_i$  is a maximum point of  $L_{d+1}$  in  $z$ , it is also a local maximum, and therefore  $z|_{v_i+} = z|_{v_i}$ . Combining the two, we get that  $z|_{v_i}$  is a codeword of the small code at  $v_i$ , namely  $z|_{v_i} \in \mathcal{C}_{v_i}^\pm$ . Thus, by Claim 1.6, there is a stabilizer  $w_i \in \text{Stab}_{v_i}^\pm$  such that  $w_i$  and  $z$  agree on the positive faces of  $v_i$ , namely  $w_i|_{v_i} = z|_{v_i}$ .

Set  $\tilde{z} \leftarrow z + \sum_i w_i$ . Since  $\sum_i w_i$  is a stabilizer, we have  $\tilde{z} \sim_{\mathcal{C}_X} z$ . In addition, for any  $j \neq i$ , it follows that  $v_j \notin w_i$ . Otherwise, there exists  $S' \subset S$  such that  $v_j \in \theta_{v_i, S'}$ , and therefore  $L_{d+1}(v_j) > L_{d+1}(v_i)$ , in contradiction to  $v_i$  and  $v_j$  being both maximum points for  $L_{d+1}$  in  $\partial^\pm z$ . Thus, we have:

$$\tilde{z}|_{v_i} = (z + \sum_i w_i)|_{v_i} = z|_{v_i} + (z + w_i)|_{v_i} = 0$$

Namely, for any  $i$ , the  $i$ th vertex  $v_i$  is not in  $\tilde{z}$ . Hence, the vertex domain of  $\tilde{z}$  is contained in the union of the vertices of  $z$  with the vertices of  $w_1, w_2, \dots, w_k$ , substituting  $v_1, v_2, \dots, v_k$ . Overall, we get:

$$\begin{aligned} \max \{L_{d+1}(u) : u \in \tilde{z}\} &\leq \max \{L_{d+1}(u) : u \in z \bigcup_i w_i / \bigcup_i v_i\} \\ &\leq \max \{L_{d+1}(u) : u \in z\} - 1 \end{aligned}$$

□

**Claim 1.8.** Let  $z$  be a finite assignment. Then there is  $\tilde{z} \sim_{C_X} z$  such that

$$\max \{L_{d+1}(u) : u \in \tilde{z}\} = \max \{L_{d+1}(u) : u \in \partial^\pm z\}$$

*Proof.* Let  $z$  be a finite assignment, and let  $k$  be an integer such that  $k \geq 1$  and

$$\max \{L_{d+1}(u) : u \in z\} = \max \{L_{d+1}(u) : u \in \partial^\pm z\} + k$$

For an integer  $l \leq k$ , consider the assignments  $\tilde{z}_0, \tilde{z}_1, \dots, \tilde{z}_l$  defined as follow:

- The first element,  $\tilde{z}_0 = z$ .
- If the  $i - 1$ th element satisfies

$$(\star) \quad \max \{L_{d+1}(u) : u \in \tilde{z}_{i-1}\} > \max \{L_{d+1}(u) : u \in \partial^\pm \tilde{z}_{i-1}\}$$

Then define the  $i$ th element,  $\tilde{z}_i$ , to be the result of applying the construction, given in Claim 1.7, on  $\tilde{z}_{i-1}$ , namely  $\partial \tilde{z}_{i+1} = \partial \tilde{z}_i$  and

$$\max \{L_{d+1}(u) : u \in \tilde{z}_i\} \leq \max \{L_{d+1}(u) : u \in \tilde{z}_{i-1}\} - 1$$

If the  $i - 1$ th element doesn't satisfies  $\star$ , then set  $l = i - 1$ .

- If the  $k$ th element was defined, set  $l = k$ .

Since  $\sim_{C_X}$  is a transitive relation, we have that for any  $i \in [l]$ ,  $z \sim_{C_X} \tilde{z}_i$ . If  $l \neq k$ , then there exists  $\tilde{z}_i$  such that  $\tilde{z}_i \sim_{C_X} z$  and  $\tilde{z}_i$  satisfies  $\star$ , namely by setting  $\tilde{z} = \tilde{z}_i$ , we get the desired. So, it is left to handle the case where  $l = k$ . In that case, set  $\tilde{z} = \tilde{z}_k$ . By Claim 1.7:

$$\max \{L_{d+1}(u) : u \in \tilde{z}\} \leq \max \{L_{d+1}(u) : u \in \partial^\pm z\}$$

Yet, since  $\tilde{z} \sim_{C_X} z$ , it follows that  $\partial \tilde{z} = \partial z$ , So:

$$\max \{L_{d+1}(u) : u \in \partial^\pm z\} = \max \{L_{d+1}(u) : u \in \partial^\pm \tilde{z}\} \leq \max \{L_{d+1}(u) : u \in z\}$$

Therefore, we have the equality:

$$\max \{L_{d+1}(u) : u \in \tilde{z}\} = \max \{L_{d+1}(u) : u \in \partial^\pm z\}$$

□

**Claim 1.9.** Let  $z$  be a finite assignment and let  $\xi$  be the result of the application of the Sweep rule on it. Then:

1. For any  $i \in [d]$ :  $\max \{L_i(v) : v \in \partial z\} = \max \{L_i(v) : v \in \partial \xi\}$
2. And for  $i = d + 1$ :  $\max \{L_{d+1}(v) : v \in \partial z\} > \max \{L_{d+1}(v) : v \in \partial \xi\} + 1$

*Proof.* First, since the Sweep rule is invariant to addition by stabilizers, for any  $\tilde{z}$  such that  $\tilde{z} \sim_{C_X} z$ , the correction applied by the Sweep rule on  $\tilde{z}$  is the same as the application on  $z$ . Denote by  $\phi$  the correction, by  $\tilde{\xi}$  the result of applying the Sweep rule on  $\tilde{z}$ , and by  $w$  the stabilizer which is the difference between  $z$  and  $\tilde{z}$ , namely:

$$\begin{aligned}\xi &= z + \phi \\ \tilde{\xi} &= \tilde{z} + \phi \\ \tilde{z} &= z + w\end{aligned}$$

The  $\partial^\pm$  boundary of  $\tilde{\xi}$  equals:

$$\partial \tilde{\xi} = \partial(\tilde{z} + \phi) = \partial(z + \phi) = \partial \xi$$

So, in total, we have that for any  $\tilde{z}$  such that  $\tilde{z} \sim_{C_X} z$ , it holds that  $\partial \tilde{z} = \partial z$  and  $\partial \tilde{\xi} = \partial \xi$ . Therefore, showing that there exists  $\tilde{z} \sim_{C_X} z$  for which the claim holds proves the claim for  $z$ . By Claim 1.8, there is  $\tilde{z} \sim_{C_X} z$  such that:

$$\max \{L_{d+1}(u) : u \in \tilde{z}\} = \max \{L_{d+1}(u) : u \in \partial^\pm \tilde{z}\}$$

For such  $\tilde{z}$ , any maximum of  $L_{d+1}$  in  $\partial \tilde{z}$ , is also a maximum in  $\tilde{z}$ . Denote by  $v_1, \dots, v_k$  the maximum points in  $\partial \tilde{z}$ . Since they are maximum points in  $\tilde{z}$ , it holds that for each  $i \in [k]$ ,  $\tilde{z}|_{v_i} = \tilde{z}|_{v_i+}$ , and hence  $\partial \tilde{z}|_{v_i} = \partial \tilde{z}|_{v_i+}$ . Since  $\tilde{z}|_{v_i+}$  is an assignment rooted at  $v_i$ , then  $v_i$  can suggest  $\tilde{z}|_{v_i+}$ , namely the condition in Sweep rule is satisfied for vertex  $v_i$ . Thus:

$$\partial \tilde{\xi}|_{v_i} = \partial \tilde{z}|_{v_i} + \partial \phi|_{v_i} = 0$$

Namely, for any  $i$ , the  $i$ th vertex  $v_i$  is not in  $\tilde{\xi}$ . For any vertex  $v$  denote by  $\phi_v$  the correction made by the vertex  $v$ , namely  $\phi = \sum_{v \in \partial \tilde{z}} \phi_v$ . Hence, the vertex domain of  $\tilde{\xi}$  is contained in the union of the vertices of  $\tilde{z}$  with the vertices of  $\cup_{v \in V} \phi_v$ , substituting maximum points  $v_1, v_2, \dots, v_k$ . Overall, we get:

$$\begin{aligned}\max \{L_{d+1}(u) : u \in \partial \tilde{\xi}\} &\leq \max \{L_{d+1}(u) : u \in \partial \tilde{z} \cup \bigcup_{v \in \partial \tilde{z}} \phi_v / \bigcup_i v_i\} \\ &\leq \max \{L_{d+1}(u) : u \in \partial \tilde{z}\} - 1\end{aligned}$$

Let  $v$  be a vertex that is a maximum of  $L_g$  in  $\partial \tilde{z}$ , maximum points of  $L_g$  do not see syndrome at the  $g$  direction. If and:

$$\phi_v = \sum \theta_{\mathbf{v}, \mathbf{S}'_i}$$

By Claim 1.3  $\partial\tilde{z}|_{gv} = 0$ , Hence:

$$\begin{aligned}\partial\phi_v|_{gv} &= \left(\partial \sum \theta_{\mathbf{v}, \mathbf{S}'_i}\right)|_{gv} \\ &= \left(\sum_{\mathbf{S}'_i} \sum_{g' \in \mathbf{S}'_i} \theta_{\mathbf{v}, \mathbf{S}'_i/g'} + \theta_{g'\mathbf{v}, \mathbf{S}'_i/g'}\right)|_{gv} \\ &= 0\end{aligned}$$

Thus if  $g \in \cup \mathbf{S}'_i$ , Then:

$$\theta_{g\mathbf{v}, \mathbf{S}'_i/g} \in \partial\phi_v|_{gv}$$

So for  $\theta_{g\mathbf{v}, \mathbf{S}'_i/g} = 0$ , it must appear at least twice in the sum, yet the left terms in the sum cannot contain it (they differ by the root vertex), and if there is a right term  $\theta_{g\mathbf{v}, \mathbf{S}'_j/g} = \theta_{g\mathbf{v}, \mathbf{S}'_i/g}$ , then it implies that  $\mathbf{S}_j = \mathbf{S}_i$ . Which is a contradiction, therefore:  $g \notin \cup \mathbf{S}'_i$  and  $gv \notin \phi_v$ .

$$\begin{aligned}\max \left\{ L_g(u) : u \in \partial\tilde{\xi} \right\} &\leq \max \left\{ L_g(u) : u \in \partial\tilde{z} \bigcup_{v \in \partial\tilde{z}} \phi_v / \bigcup_i v_i \right\} \\ &\leq \max \left\{ L_g(u) : u \in \partial\tilde{z} \right\} + 1 - 1\end{aligned}$$

□